figures-referenced

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Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
library(lme4)
library(ggplot2)
library(patchwork)
library(ggpmisc)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

gradient_it <- read_csv(here("data", "it_gradient_data.csv"))
```

```r
metadata <- read_csv(here("data", "metadata.csv"))

site_order <- c("tc_us", "tc_mid", "tc_ds", "tc_1", "tc_15", "tc_2", "tc_3")

trail_creek_isco <- compiled_isotope |>
  filter(site %in% c("utc_isco", "tc_isco"))

trail_creek <- compiled_isotope |>
  filter(site %in% site_order) |>
  mutate(site = factor(site, levels = site_order)) |>
  arrange(site)

coal_creek <- compiled_isotope |>
 filter(location %in% c("cc"))

italian_creek <- compiled_isotope |>
  filter(location == "it")

trail_creek_isco <- read_csv(here("data", "trail_creek_compiled_ISCO.csv")) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stdev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stdev) |>
  mutate(date = mdy(date))# telling r the date format

utc_isco <- trail_creek_isco |>
  filter(site == "utc_isco")

tc_isco <- trail_creek_isco |>
   filter(site == "tc_isco")


# Downstream Trail Creek
tc_metrics <- read_csv(here("data", "TrailCreek_2026-07-23-15.csv"), show_col_types = FALSE)
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
         dp = differential_pressure_k_pa,
         ap = absolute_pressure_k_pa,
         temp = temperature_c,
         water_level = water_level_m,
```

```
        bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
        date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_metrics <- read_csv(here("data", "UTC_2026-07-23-15.csv"), show_col_types = FALSE) |> # i
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt,  # renaming columns to be shorter
        dp = differential_pressure_k_pa,
        ap = absolute_pressure_k_pa,
        temp = temperature_c,
        water_level = water_level_m,
        bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
        date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE),  # averaging the data based on the day
            avg_ap = mean(ap, na.rm = TRUE),
            avg_temp = mean(temp, na.rm = TRUE),
            avg_water_level = mean(water_level, na.rm = TRUE),
            avg_bp = mean(bp, na.rm = TRUE),
            total_readings = n(), # the amount of data it used to average
            .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

tc_utc_metrics <- merge(tc_metrics, utc_metrics, all = TRUE) # merging the tc and utc data

isco_data <- merge(
  x = tc_utc_metrics,
  y = compiled_isotope,
  by.x = c("site", "date"), # columns in tc_utc_data
  by.y = c("location", "date") # matching columns in compiled_isotope
)
```

```r
discharge_tc_isco <- read_csv(here("data", "discharge_tc_isco.csv"))
```

Figures

Figure 1: Time-Series of Line-Conditioned Excess (lc-excess) of ISCO's

```r
ggplot(data = trail_creek_isco,
       aes(x = date,
           y = lc_excess,
           color = site,
           group = site)) +
  geom_hline(yintercept = 0,
             linetype = "dashed",
             color = "gray50",
             size = 0.8) +
  geom_point(size = 2.5,
             alpha = 0.8) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +

  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\", \"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "top", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.5) +
  scale_color_manual(
    values = c("tc_isco" = "dodgerblue", "utc_isco" = "orangered2"),
    labels = c( "tc_isco" = "Trail Creek", "utc_isco" = "Upper Trail Creek")
  ) +

  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
       subtitle = "Values below 0 indicate potential evaporative enrichment",
       x = "Sampling Date",
       y = "lc-excess (\u2030)",
       color = "Site") +
  ylim(-8,5) +
  theme_bw() +
```

```
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```

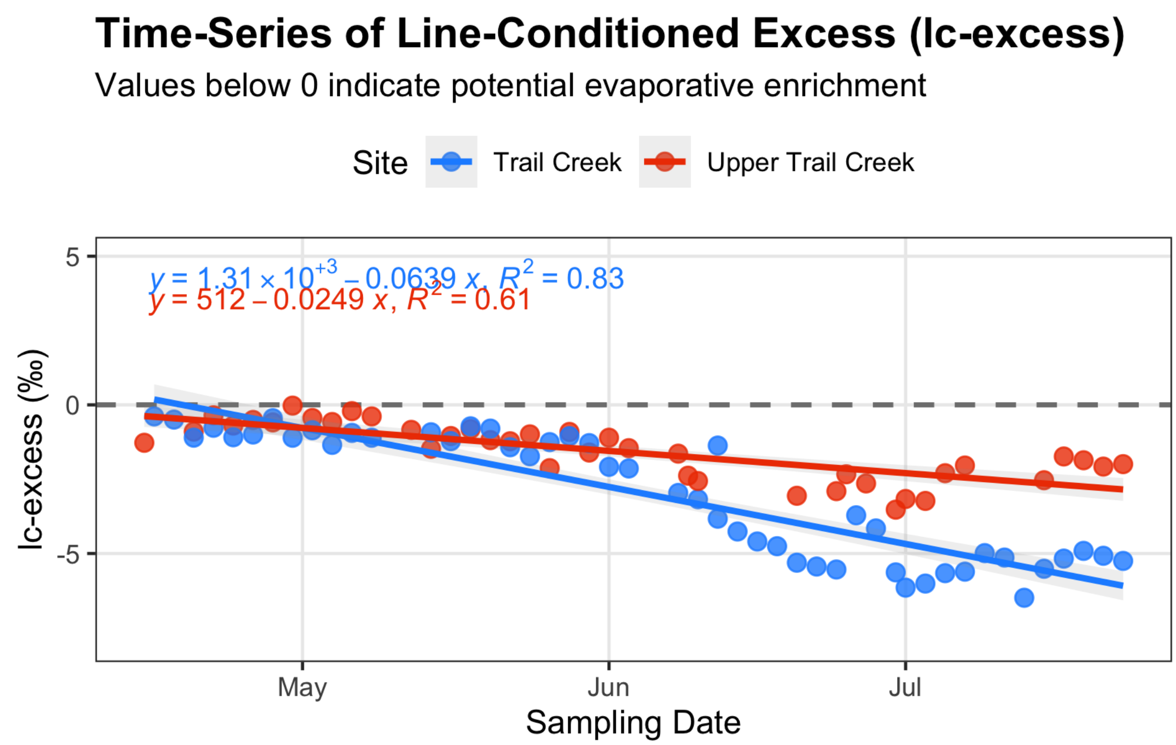


Figure 1.1: Sampling Site and lc-excess (Trail Creek)

```
plot1_1 <- ggplot(data = trail_creek, aes(x = site, y = lc_excess, color = factor(date), grou
  # reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  # adds a linear regression line for each date group
  geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\", \"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "bottom", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.0) +
```

```
  # points for each site's value
  geom_point(size = 2.0, alpha = 0.8) +
  labs(
    title = "Trail Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\u2030)",
    color = "Sampling Date"
  ) +
  scale_color_manual(values = c("2026-06-24" = "royalblue3", "2026-06-30" = "royalblue1", "20
  scale_x_discrete(labels = c("tc_us" = "TC-US", "tc_mid" = "TC-MID", "tc_ds" = "TC-DS", "tc_
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
plot1_1
```

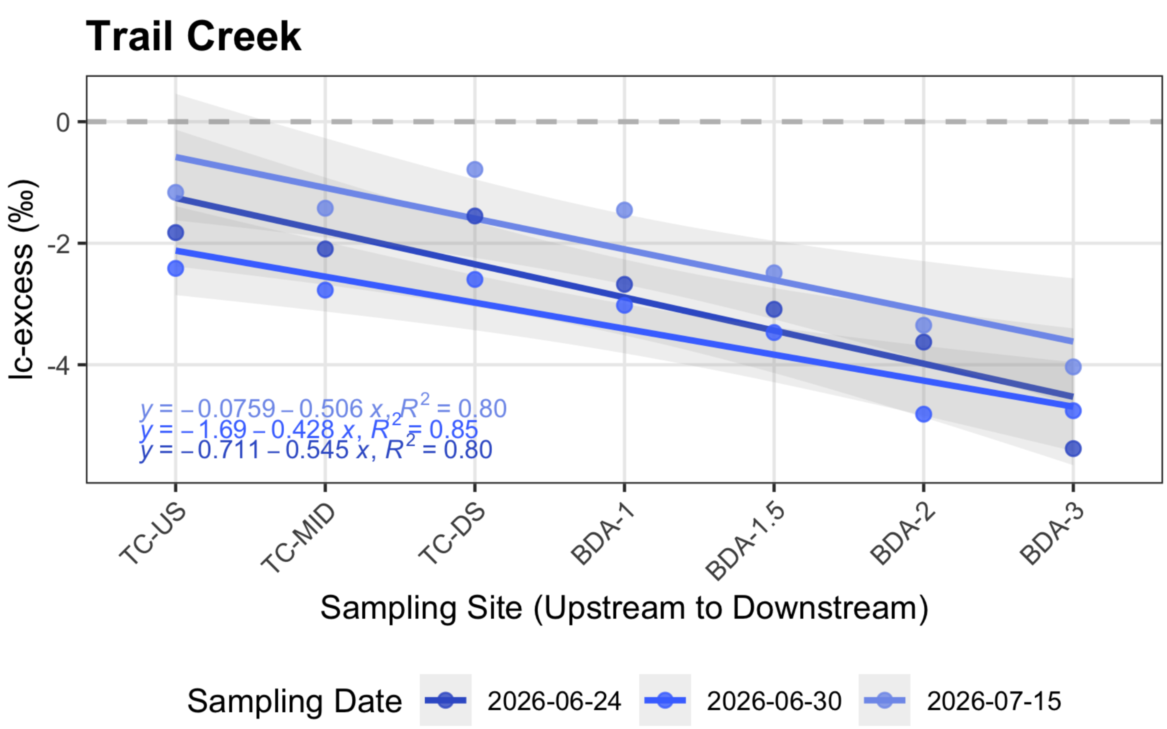


Figure 1.2: Sampling Site and lc-excess (Coal Creek)

```r
plot1_2 <- ggplot(data = coal_creek, aes(x = site, y = lc_excess, color = factor(date), grou
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, linewidth = 1, alpha = 0.15) +
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\", \"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left",
    label.y = "bottom",
    vstep = 0.05,
    size = 3.0) +
  geom_point(size = 2.0, alpha = 0.8) +
  scale_color_manual(values = c("2026-06-23" = "royalblue3", "2026-07-07" = "royalblue1")) +
  scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coal_
  labs(
    title = "Coal Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\u2030)",
    color = "Sampling Date"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
plot1_2
```

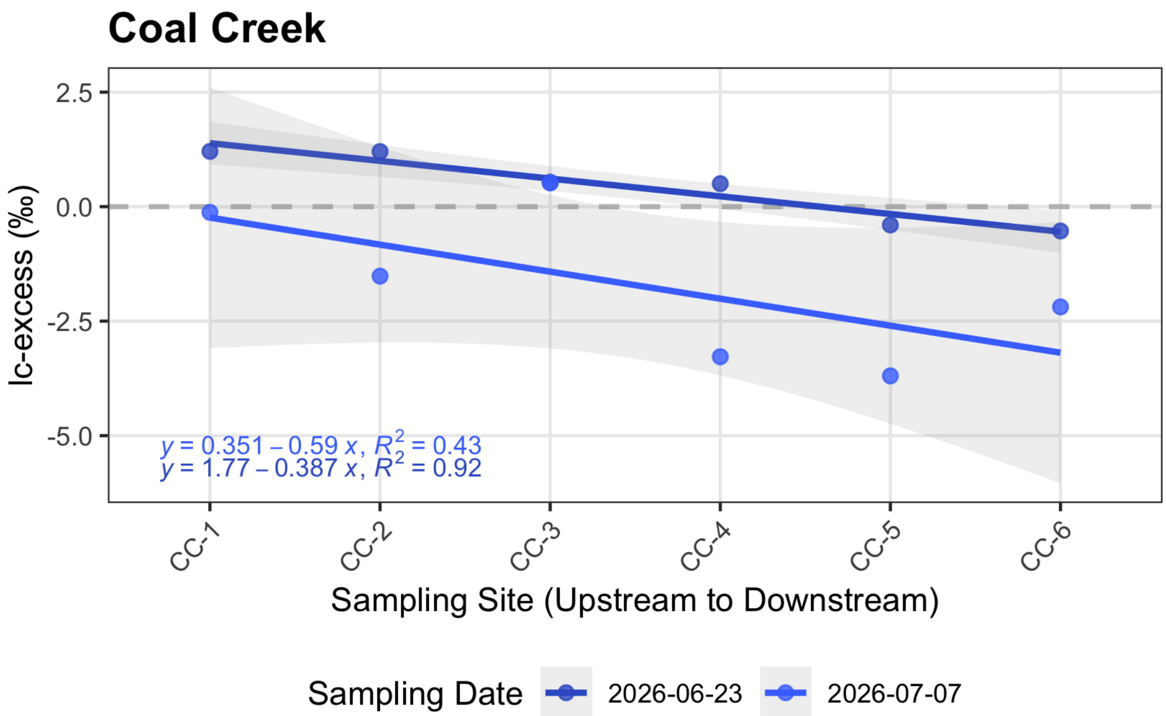
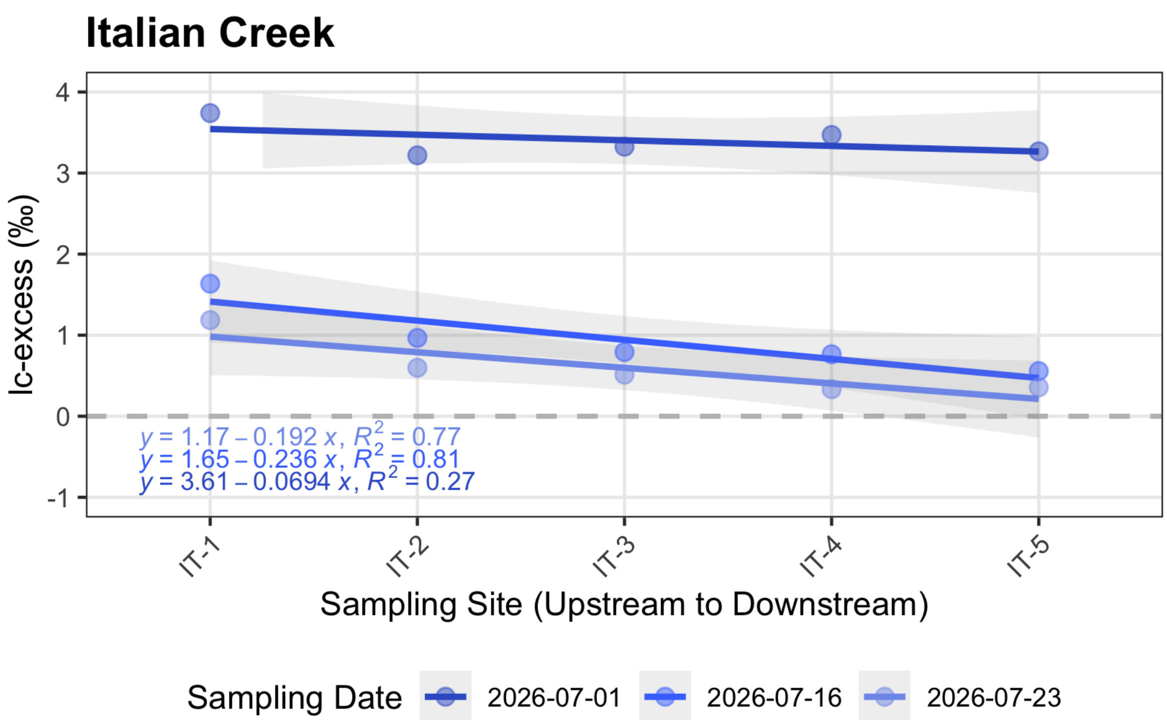


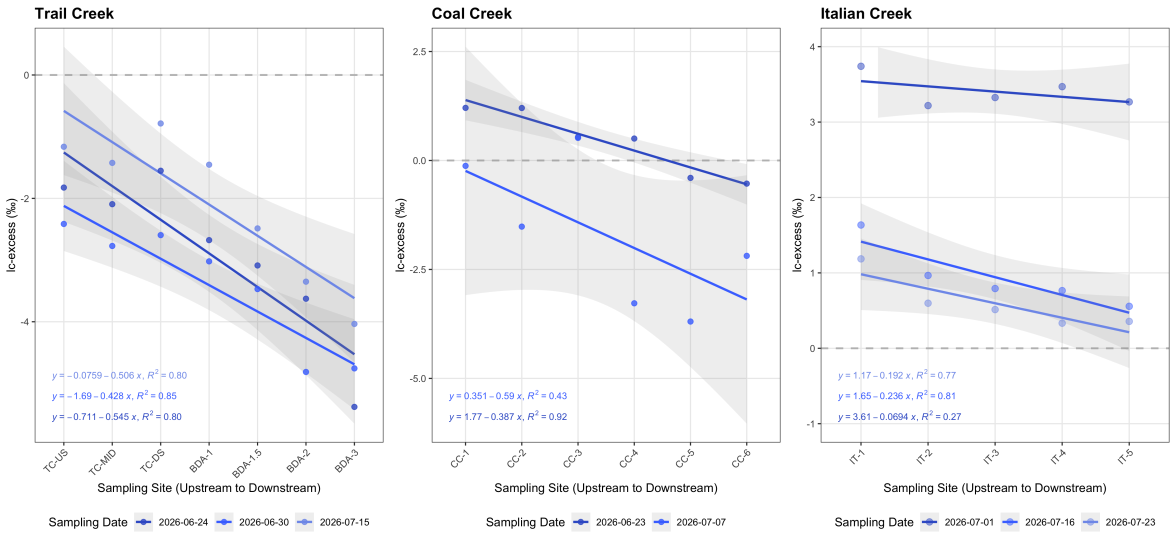
Figure 1.3: Sampling Site and lc-excess (Italian Creek)

```
plot1_3 <- ggplot(data = italian_creek, aes(x = site, y = lc_excess, color = factor(date), g
  # reference line at 0
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray", linewidth = 0.8) +
  # adds a linear regression line for each date group
  geom_smooth(method = "lm", se = TRUE, linewidth = 1, alpha = 0.15) +
  # adds the equation and R-squared text automatically for each date line
  stat_poly_eq(
    aes(label = paste(after_stat(eq.label), after_stat(rr.label), sep = "*\", \"*")),
    formula = y ~ x,
    parse = TRUE,
    label.x = "left", # positions the labels horizontally
    label.y = "bottom", # positions the labels vertically
    vstep = 0.05, # spacing between the text lines so they do not overlap
    size = 3.0) +
    ylim(-1,4) +
  # points for each site's value
  geom_point(size = 2.5, alpha = 0.5) +
  scale_color_manual(values = c("2026-07-01" = "royalblue3", "2026-07-16" = "royalblue1", "2
  scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-
```

```r
  labs(
    title = "Italian Creek",
    x = "Sampling Site (Upstream to Downstream)",
    y = "lc-excess (\u2030)",
    color = "Sampling Date"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
plot1_3
```

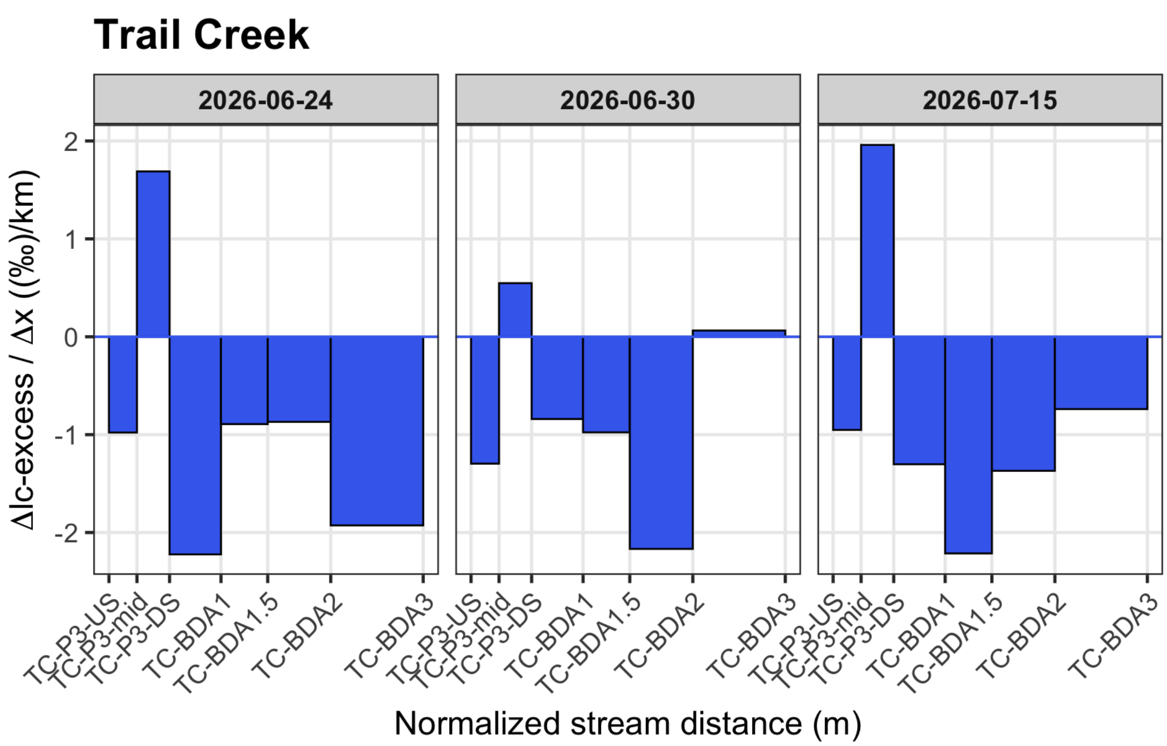



```r
plot1_1 + plot1_2 + plot1_3
```



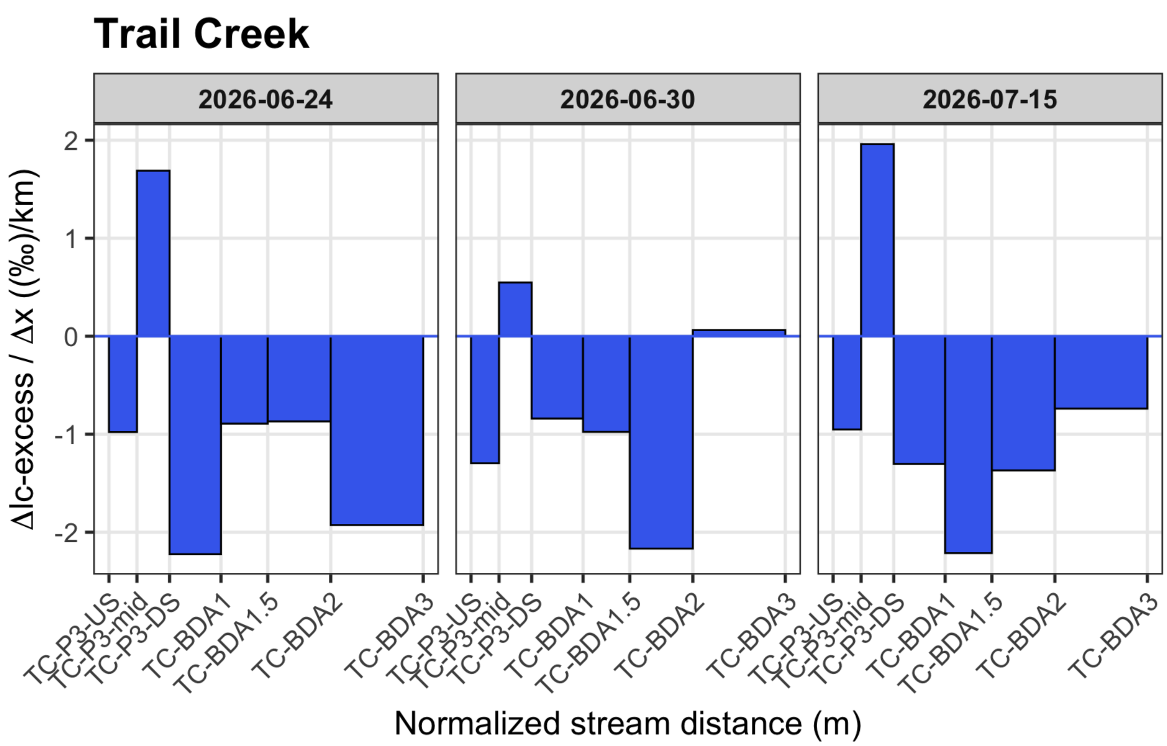

```r
# Spatial variations in Δlc-excess/Δx across sampling dates
# Buildings a look up table of stream distances Trail Creek
distances <- tibble(site = c("TC-P3-US", "TC-P3-mid", "TC-P3-DS", "TC-BDA1",
                             "TC-BDA1.5", "TC-BDA2", "TC-BDA3"),
                    dist_m = c(57.6, 333, 654, 1160, 1620, 2240, 3150)) |>
  mutate(xmax = dist_m,
         xmin = lag(dist_m, default = 0)) # previous site's distance, 0 for first

# Creating a filtered data frame for lc-excess
gradient_tc_clean <- gradient_tc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc
  pivot_longer( cols = c(`Δlc-excess/Δx_6-24`, `Δlc-excess/Δx_6-30`, `Δlc-excess/Δx_7-15`),
                names_to = "date",
                values_to = "gradient") |>
  mutate(date = case_when(date == "Δlc-excess/Δx_6-24" ~ "2026-06-24",
                          date == "Δlc-excess/Δx_6-30" ~ "2026-06-30",
                          date == "Δlc-excess/Δx_7-15" ~ "2026-07-15"),
         gradient = as.numeric(gradient)) |>
  left_join(distances %>% select(site, xmin, xmax), by = "site") |>
  mutate(site = factor(site, levels = distances$site)) |>
  filter(!is.na(gradient))

# Gradient plot
plot1 <- ggplot(gradient_tc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
```

```r
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances$dist_m,
    labels = distances$site) +
  labs(title = "Trail Creek",
       x = "Normalized stream distance (m)",
       y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        strip.text = element_text(face = "bold"),
        axis.text.x = element_text(angle = 45, hjust = 1))
plot1
```

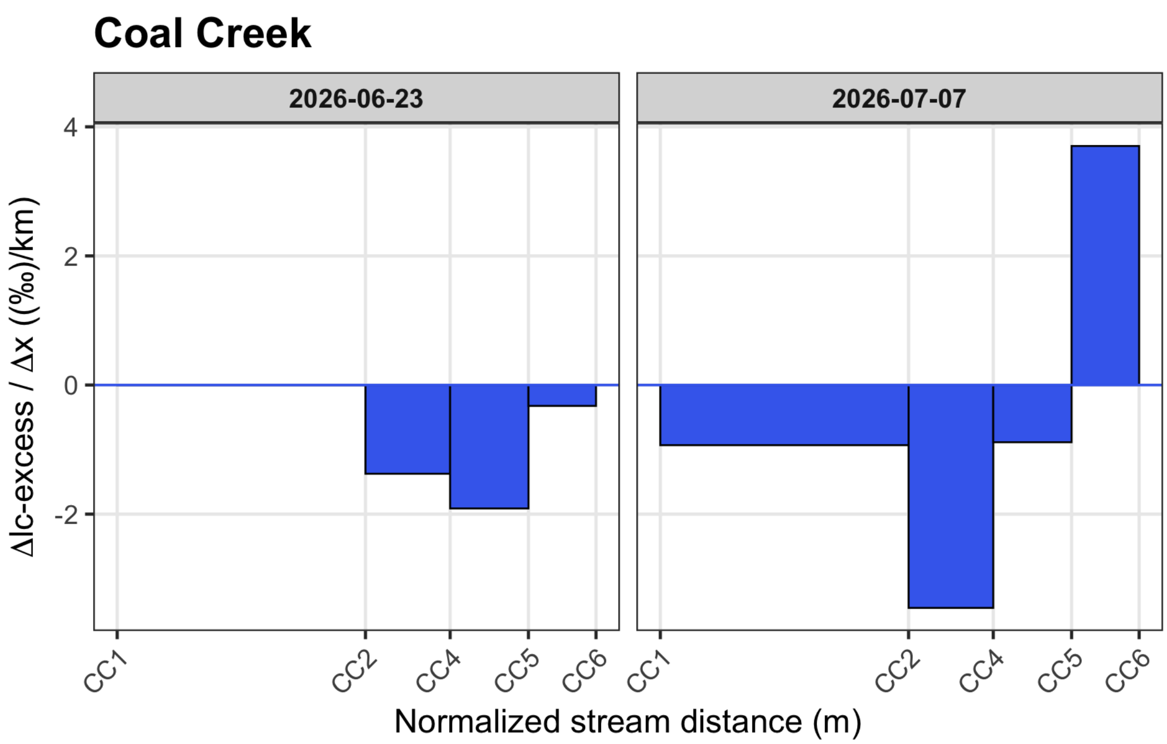



```r
# Buildings a look up table of stream distances Coal Creek
distances_cc <- tibble(
  site = c("CC1", "CC2", "CC4", "CC5", "CC6"),
  dist_m = c(0, 1496, 2006, 2478, 2885)) |>
  mutate(xmax = dist_m,
```

```
         xmin = lag(dist_m, default = 0))

# Creating a filtered data frame for lc-excess
gradient_cc_clean <- gradient_cc |>
  select(site, normalized_stream_distance_m, `Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`) |>
  pivot_longer(cols = c(`Δlc-excess/Δx_6-23`, `Δlc-excess/Δx_7-7`),
    names_to = "date",
    values_to = "gradient") |>
  mutate(date = case_when(
    date == "Δlc-excess/Δx_6-23" ~ "2026-06-23",
    date == "Δlc-excess/Δx_7-7" ~ "2026-07-07"),
    gradient = as.numeric(gradient)) |>
  left_join(distances_cc |> select(site, xmin, xmax),
            by = "site") |>
  mutate(site = factor(site, levels = distances_cc$site)) |>
  filter(!is.na(gradient))

plot1
```



```r
# Gradient plot
plot2 <- ggplot(gradient_cc_clean) +
  geom_rect(aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
            fill = "royalblue2", color = "black", linewidth = 0.3) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_cc$dist_m,
    labels = distances_cc$site) +
  labs(
    title = "Coal Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1))

plot2
```




```
library(tidyverse)

# 1. Build the look-up table for Italian Creek
distances_it <- tibble(
  site = c("IT1", "IT2", "IT3", "IT4", "IT5"),
  dist_m = c(0, 793.37, 1665.83, 2385.13, 2963.62) # Keeps a clean width layout like plot1
) |>
  mutate(
    xmax = dist_m,
    xmin = lag(dist_m, default = 0)
  )

# 2. Creating a filtered data frame for Italian Creek (Fixed Column Names)
gradient_it_clean <- gradient_it |>
  select(site, normalized_stream_distance_m, `d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23
  pivot_longer(
    cols = c(`d_lc/d_x_7_1`, `d_lc/d_x_7_16`, `d_lc/d_x_7_23`),
    names_to = "date",
    values_to = "gradient"
  ) |>
```

```
  mutate(
    date = case_when(
      date == "d_lc/d_x_7_1"  ~ "2026-07-01",
      date == "d_lc/d_x_7_16" ~ "2026-07-16",
      date == "d_lc/d_x_7_23" ~ "2026-07-23"
    ),
    gradient = as.numeric(gradient)
  ) |>
left_join(distances_it |> select(site, xmin, xmax),
            by = "site") |>
  mutate(site = factor(site, levels = distances_it$site)) |> # FIXED: Uses distances_it inste
  filter(!is.na(gradient))

# 3. Gradient plot for Italian Creek
plot2_5 <- ggplot(gradient_it_clean) +
  geom_rect(
    aes(xmin = xmin, xmax = xmax, ymin = 0, ymax = gradient),
    fill = "royalblue2", color = "black", linewidth = 0.3
  ) +
  geom_hline(yintercept = 0, color = "royalblue2", linewidth = 0.4) +
  facet_wrap(~ date, nrow = 1) +
  scale_x_continuous(
    breaks = distances_it$dist_m,
    labels = distances_it$site
  ) +
  labs(
    title = "Italian Creek",
    x = "Normalized stream distance (m)",
    y = expression(paste(Delta, "lc-excess / ", Delta, "x ((\u2030)/km)"))
  ) +
  theme_bw(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    panel.grid.minor = element_blank(),
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

plot2_5
```

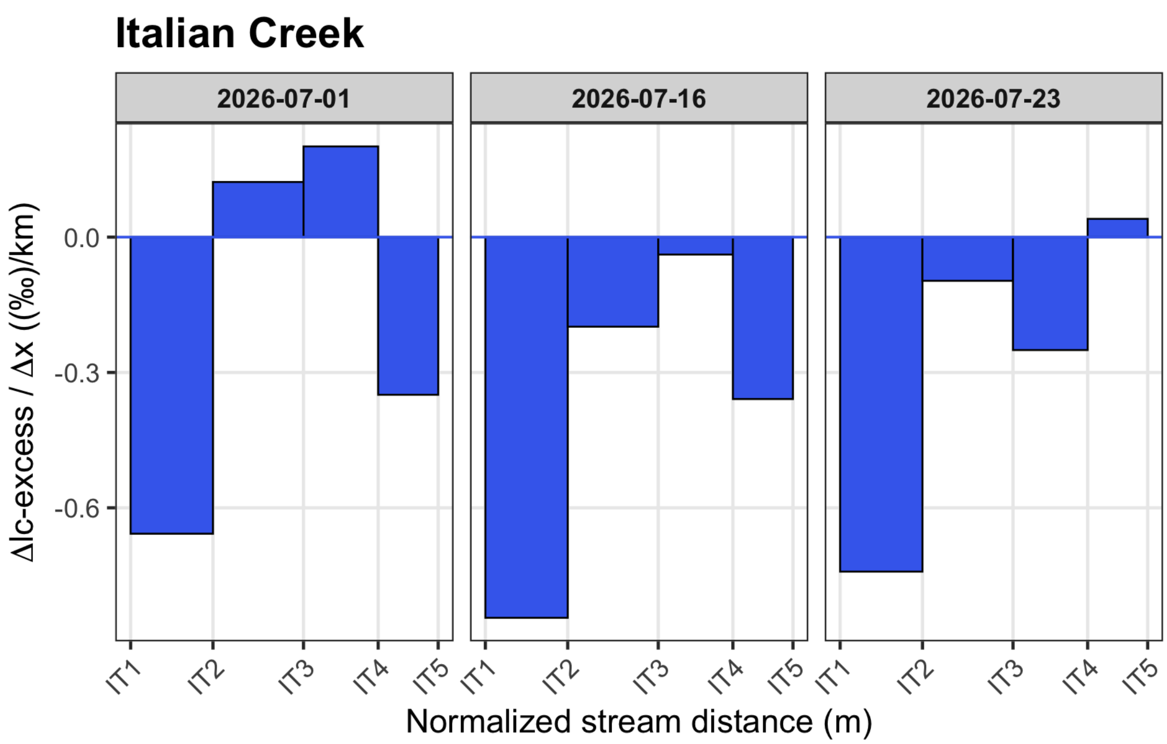
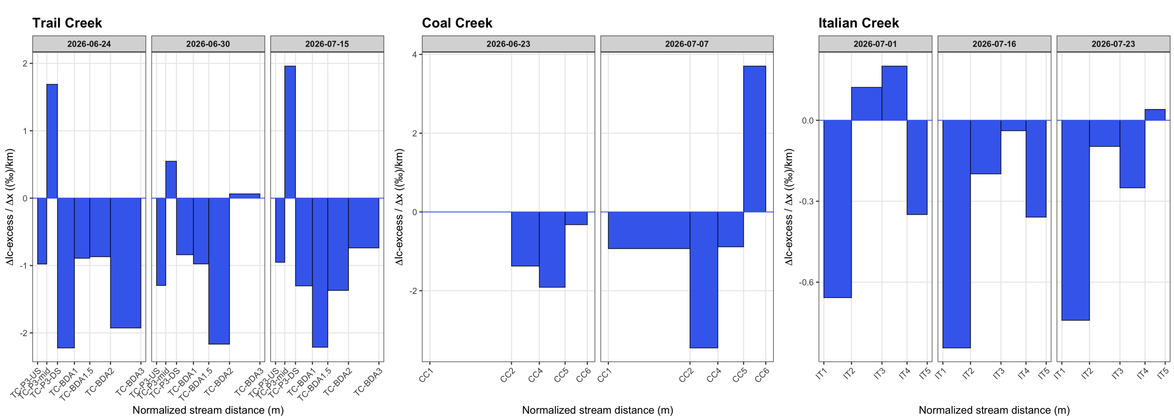


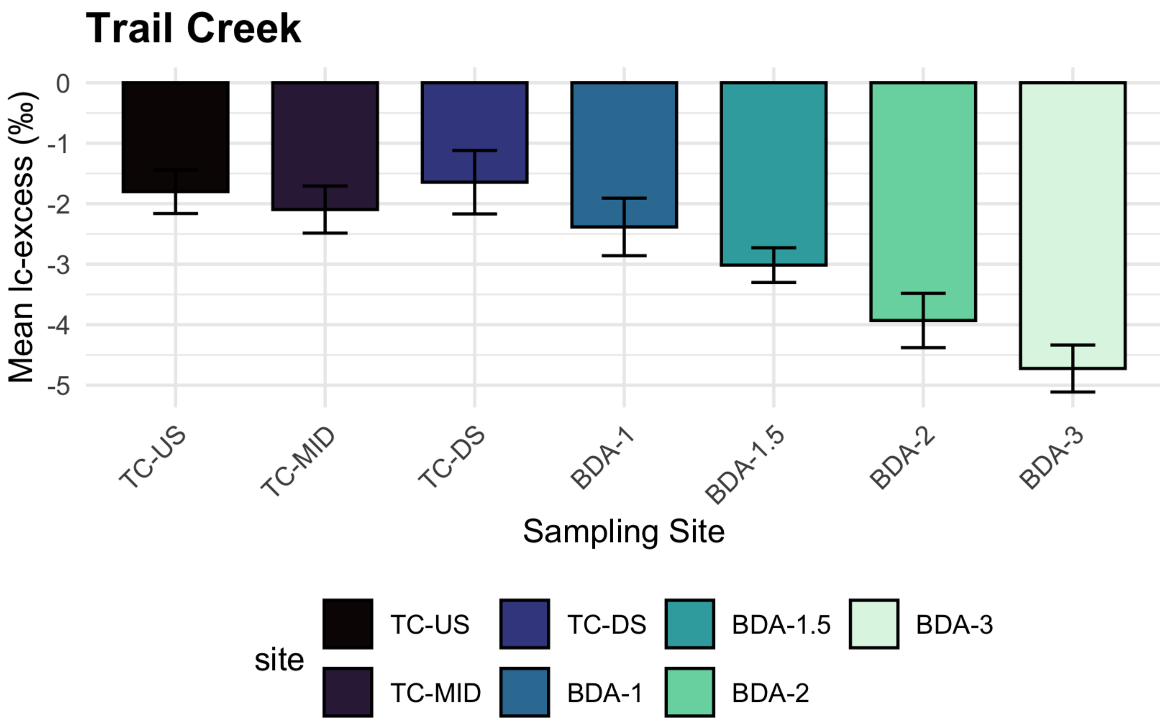
Figure 2:

```
combined_stacked <- plot1 + plot2 + plot2_5

combined_stacked # print the graph
```



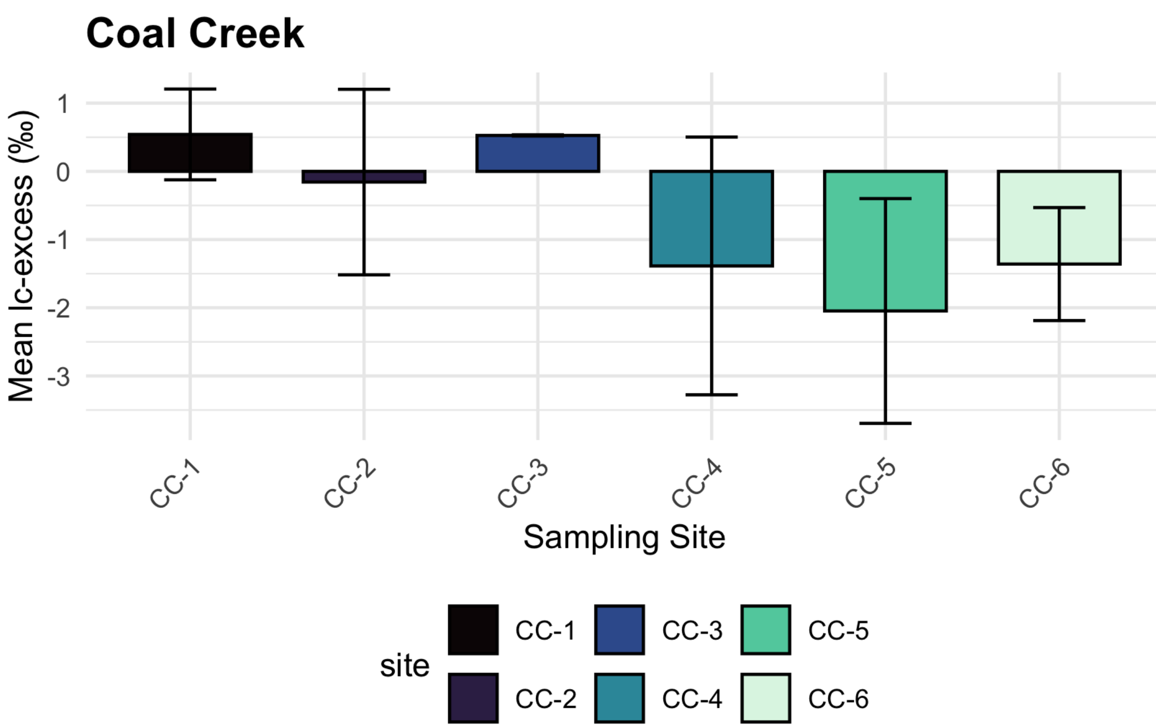

```
# averaging the lc-excess and including standard error
delta_summary <- trail_creek |>
```

```r
  mutate(site = factor(site, levels = site_order)) |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop")

# creating a plot to show the mean lc-excess of each site (trail creek)
plot3 <- ggplot(delta_summary, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3,
                color = "black") +
  scale_x_discrete(labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5", "tc_2" = "BDA-2",
              "tc_3" = "BDA-3", "tc_ds" = "TC-DS", "tc_mid" = "TC-MID", "tc_us" = "TC-US")) +
  scale_fill_viridis_d(option ="G", labels = c("tc_1" = "BDA-1", "tc_15" = "BDA-1.5", "tc_2"
              "tc_3" = "BDA-3", "tc_ds" = "TC-DS", "tc_mid" = "TC-MID", "tc_us" = "TC-US")) +
  theme_minimal() +
  labs(title = "Trail Creek",
       x = "Sampling Site",
       y = "Mean lc-excess (\u2030)") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1), # tilt labels
        legend.position = "bottom")

plot3
```




```
# averaging the lc-excess and including standard error
delta_sum_cc <- coal_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )


# creating a plot to show the mean lc-excess of each site (coal creek)
plot4 <- ggplot(delta_sum_cc, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
  scale_fill_viridis_d(option = "G", labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3
    scale_x_discrete(labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3" = "CC-3", "coa
  theme_minimal() +
  labs(title = "Coal Creek",
       x = "Sampling Site",
       y = "Mean lc-excess (\u2030)") +
```

```r
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom")
plot4
```




```r
# averaging the lc-excess and including standard error
delta_summary_it <- italian_creek |>
  group_by(site) |>
  summarise(
    mean_lc = mean(lc_excess, na.rm = TRUE),
    se_lc = sd(lc_excess, na.rm = TRUE) / sqrt(n()),
    .groups = "drop"
  )


# creating a plot to show the mean lc-excess of each site (italian creek)
plot5 <- ggplot(delta_summary_it, aes(x = site, y = mean_lc, fill = site)) +
  geom_col(color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = mean_lc - se_lc, ymax = mean_lc + se_lc),
                width = 0.3, color = "black") +
```

```
  scale_fill_viridis_d(option = "G", labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", '
  scale_x_discrete(labels = c("italian_1" = "IT-1", "italian_2" = "IT-2", "italian_3" = "IT-3
  theme_minimal() +
  labs(title = "Italian Creek",
       x = "Sampling Site",
       y = "Mean lc-excess (\u2030)") +
  theme(plot.title = element_text(face = "bold", size = 14),
        axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "bottom")

plot5
```

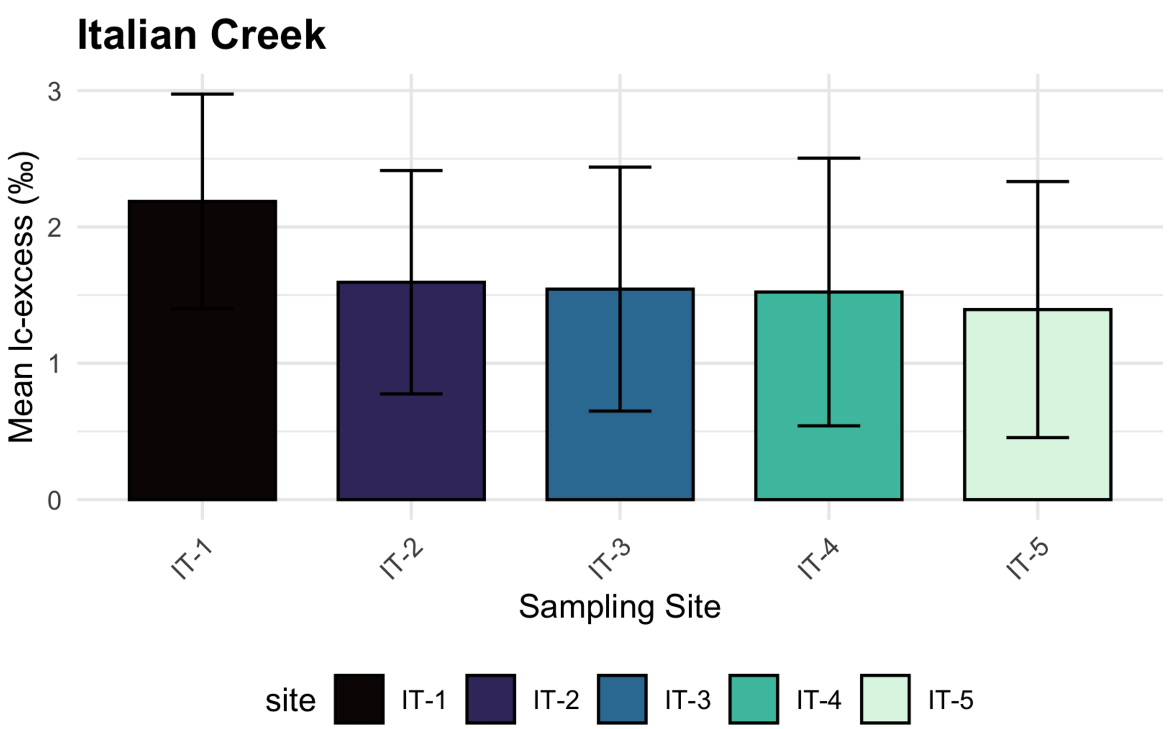


Figure 3:

```
combined_stacked_2 <- plot3 + plot4 + plot5

combined_stacked_2 # print the graph
```

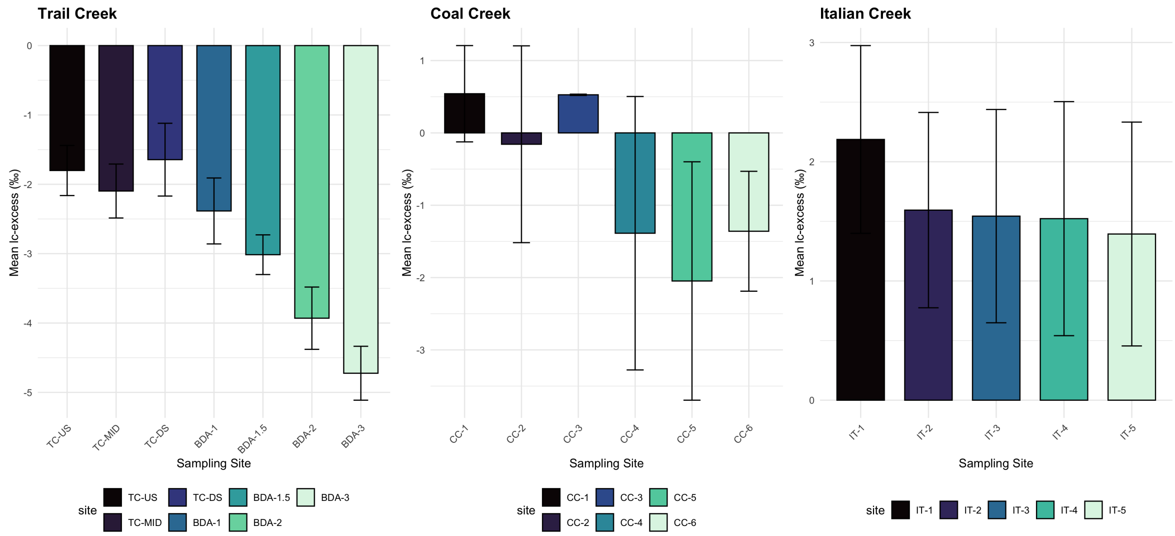
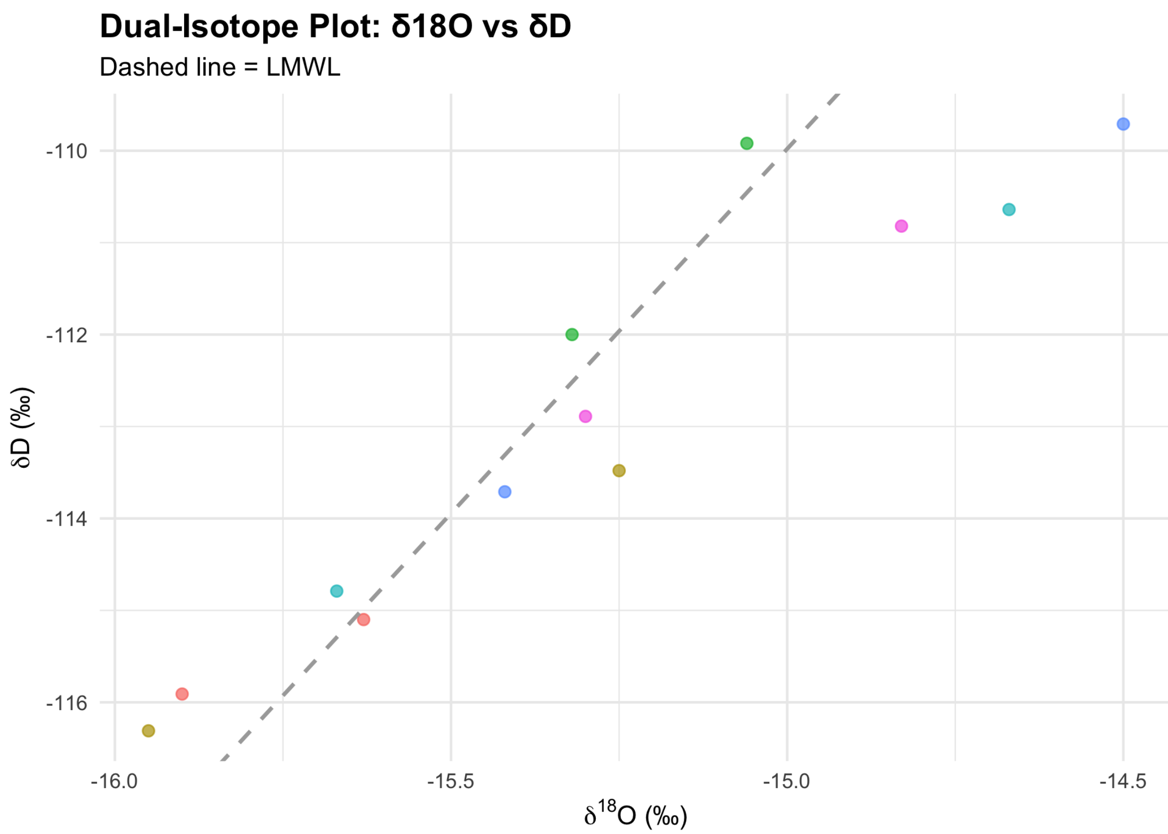
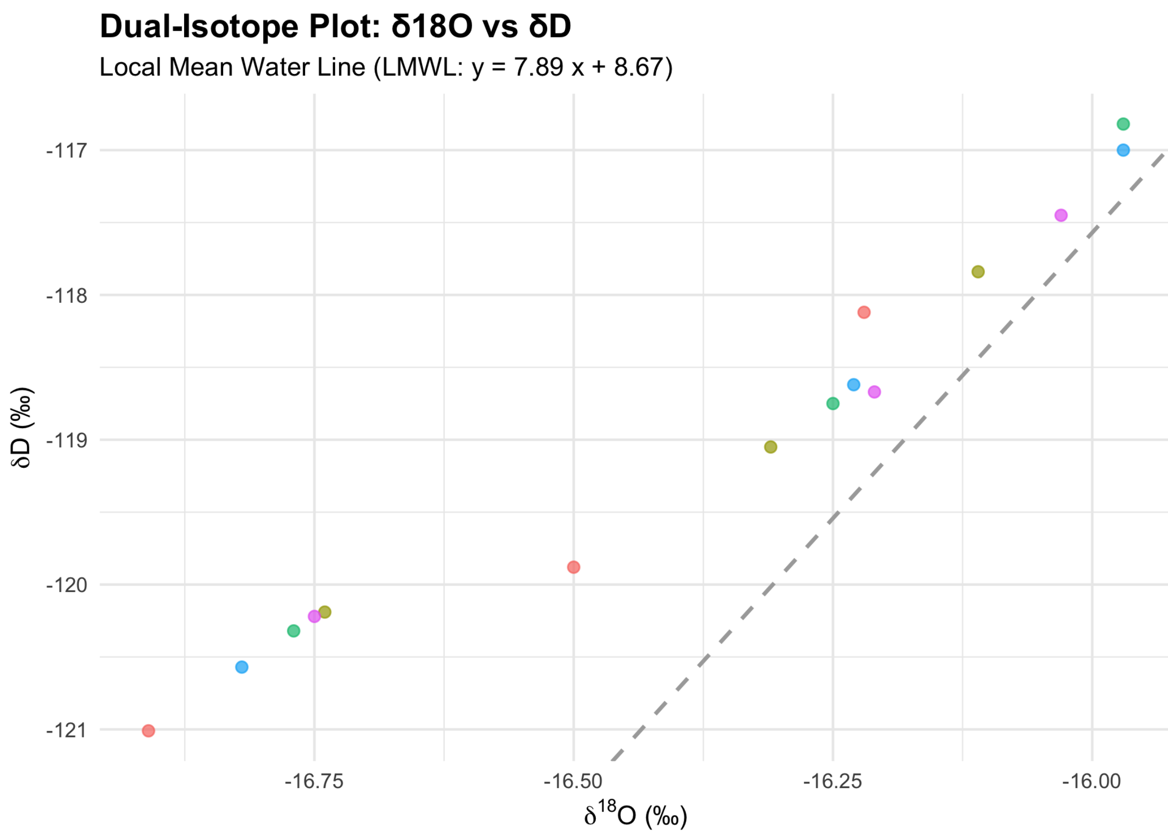


Figure 4:

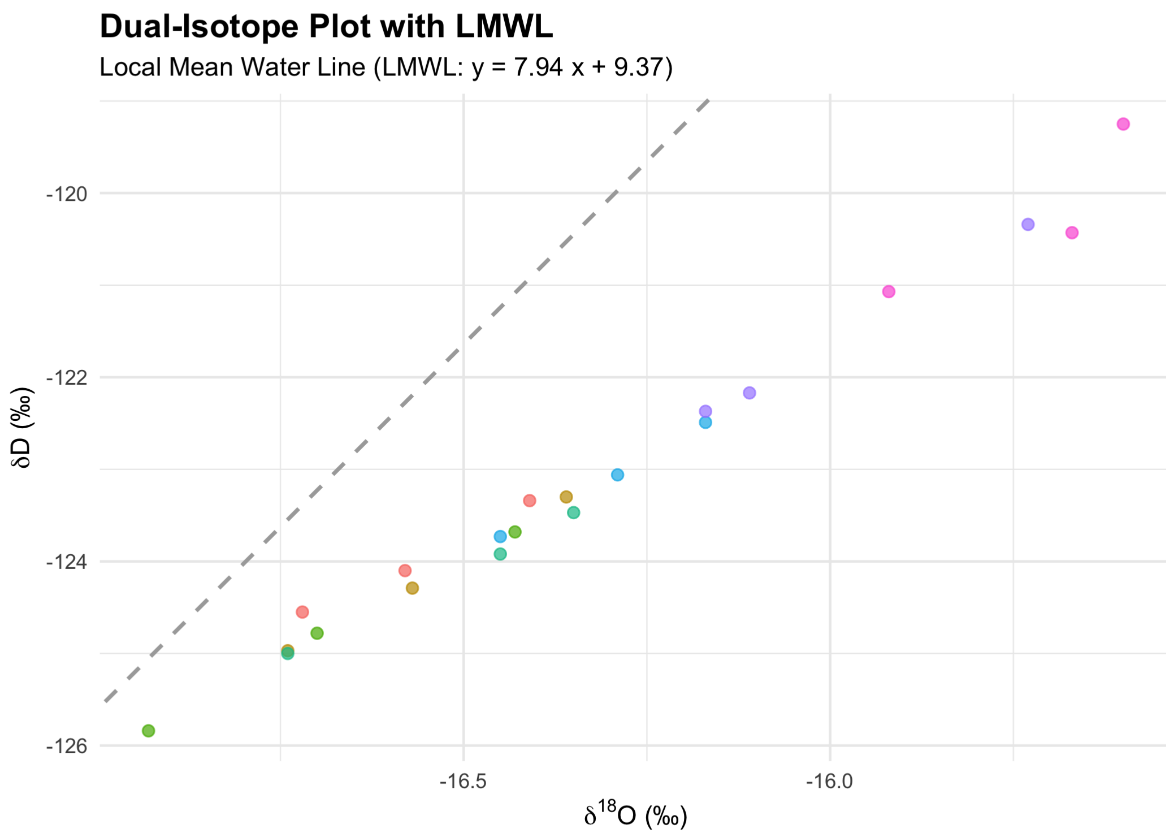
```
# Coal Creek LMWL calculated from OPIC y = 7.93 x + 8.97
ggplot(coal_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
 geom_abline(intercept = 8.97, slope = 7.93, linetype = "dashed", color = "darkgray", linewi
  labs(
    title = "Dual-Isotope Plot: \u03b4180 vs \u03b4D",
    subtitle = "Dashed line = LMWL",
    x = expression(paste(delta^{18}, "O (\u2030)")),
    y = expression(paste(delta, "D (\u2030)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```



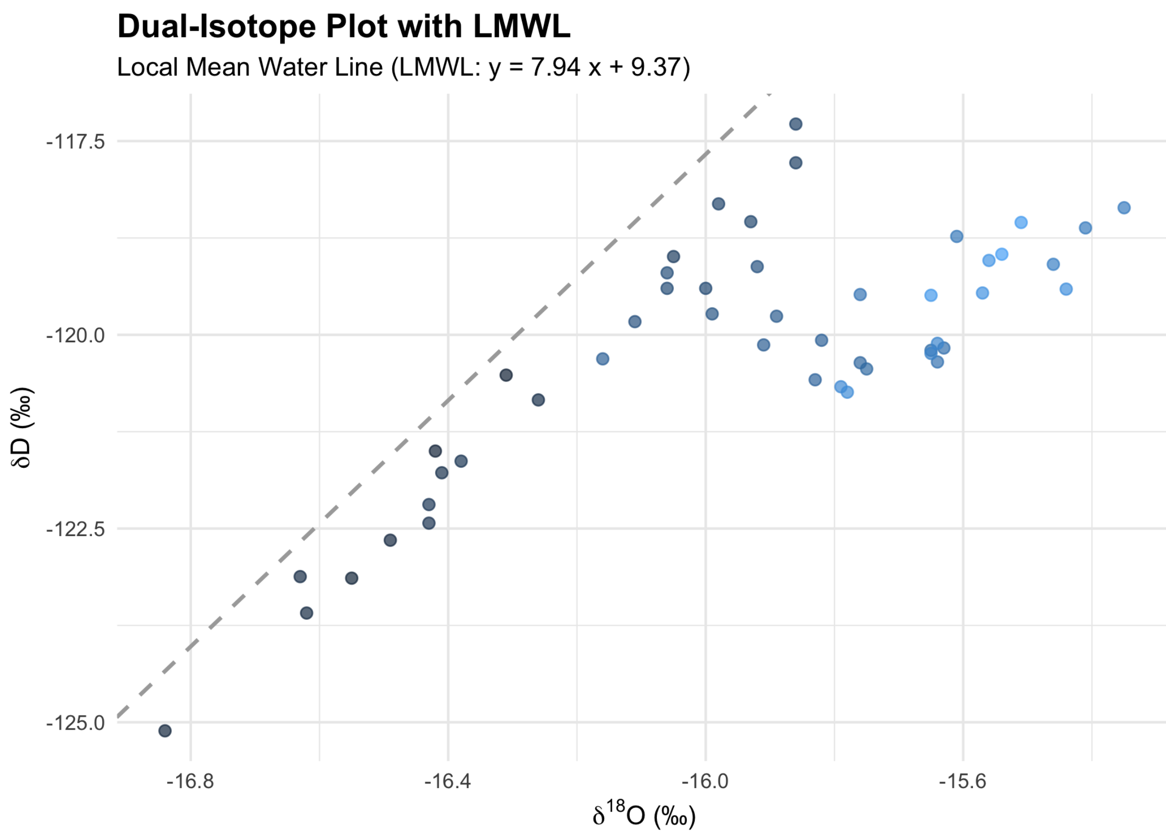

```r
# Italian Creek LMWL calculated from OPIC y = 7.89 x + 8.67

ggplot(italian_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 8.67, slope = 7.89, linetype = "dashed", color = "darkgray", linew
  labs(
    title = "Dual-Isotope Plot: \u03b4180 vs \u03b4D",
    subtitle = "Local Mean Water Line (LMWL: y = 7.89 x + 8.67)",
    x = expression(paste(delta^{18}, "O (\u2030)")),
    y = expression(paste(delta, "D (\u2030)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```



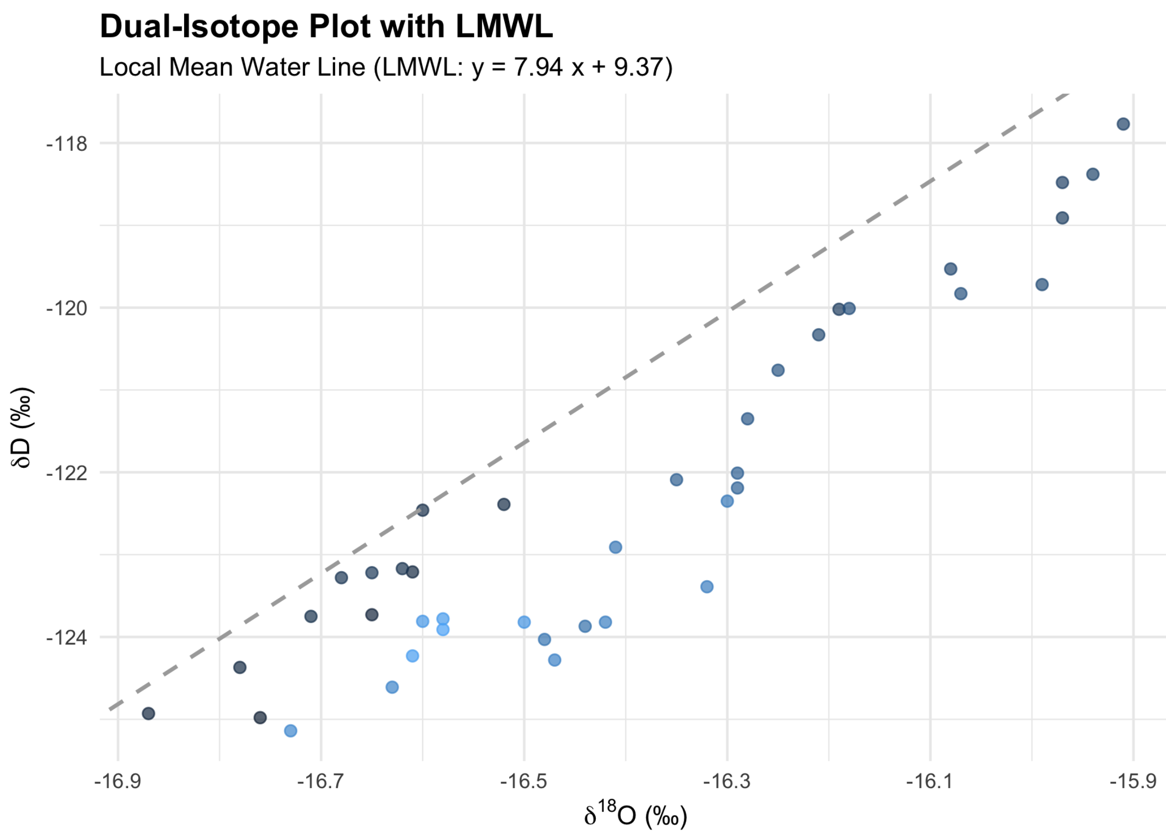

```
# Trail Creek River Reach LMWL calculated from OPIC y = 7.94 x + 9.37
ggplot(trail_creek, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linew
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL: y = 7.94 x + 9.37)",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
   theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```




```
# TC ISCO LMWL calculated from OPIC y = 7.94 x + 9.37
ggplot(tc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linew
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL: y = 7.94 x + 9.37)",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
   theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```




```
# UTC ISCO LMWL calculated from OPIC y = 7.94 x + 9.37
ggplot(utc_isco, aes(x = delta_o, y = delta_d, color = date)) +
  geom_point(alpha = 0.7, size = 2) +
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linew
  labs(
    title = "Dual-Isotope Plot with LMWL",
    subtitle = "Local Mean Water Line (LMWL: y = 7.94 x + 9.37)",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
   theme(plot.title = element_text(face = "bold", size = 14), legend.position = "none")
```



Discharge

```r
model_discharge_tc <- lm(lc_excess ~ discharge, data = discharge_tc_isco)
summary(model_discharge_tc)
```

```
Call:
lm(formula = lc_excess ~ discharge, data = discharge_tc_isco)

Residuals:
     Min       1Q   Median       3Q      Max 
-1.13810 -0.33678 -0.07501  0.29166  1.00884 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   -5.986      0.145  -41.28   <2e-16 ***
discharge     42.209      1.719   24.55   <2e-16 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5488 on 44 degrees of freedom
Multiple R-squared:  0.932, Adjusted R-squared:  0.9304
F-statistic: 602.9 on 1 and 44 DF,  p-value: < 2.2e-16
```

```
ggplot(discharge_tc_isco, aes(x = discharge, y = lc_excess)) +
  geom_point(size = 2.5, alpha = 0.7, color = "#2a78d6") +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE, color = "black", linetype = "dashed"
  stat_poly_eq(
    use_label(c("eq", "R2", "P")),
    formula = y ~ x,
    label.x = "left", label.y = "top",
    size = 3.5
  ) +
  labs(title = "TC-ISCO: Discharge vs. lc-excess",
       x = expression(paste("Discharge (cfs)")),
       y = "lc-excess (\u2030)") +
  theme_bw()
```